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Journal Section

Evaluating AI Counseling in Japanese: Counselor, Client, and Evaluator Roles Assessed by Motivational Interviewing Criteria

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This study provides the first evaluation of large language model (LLM) performance across three counseling roles in Japanese-language therapeutic contexts. We assessed counselor AI systems (GPT-4-turbo with zero-shot prompting or Structured Multi-step Dialogue Prompts [SMDP], Claude-3-Opus-SMDP), client AI simulations, and evaluation AI systems (o3, Claude-3.7-Sonnet, Gemini-2.5-pro). Human experts (n = 15) evaluated AI-generated dialogues using the Motivational Interviewing Treatment Integrity (MITI) Coding Manual 4.2.1. SMDP implementation significantly enhanced counselor AI performance across all MITI ratings compared with zero-shot prompting, with no differences between GPT-SMDP and Opus-SMDP. Evaluation AIs performed comparably to humans for Cultivating Change Talk but overestimated Softening Sustain Talk and overall quality metrics. Model-specific biases emerged: Gemini emphasized power-sharing, o3 focused on technical proficiency, and Sonnet prioritized emotional expression. Client AI simulations exhibited limited emotional range and unnaturally high compliance, requiring enhanced realism. These findings establish

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benchmarks for AI-assisted counseling in non-English contexts and identify critical areas for improvement through advanced prompt engineering, retrieval-augmented generation, and targeted fine-tuning, informing culturally sensitive AI mental health tools.

KEYWORDS
artificial intelligence, counseling, prompt engineering, evaluation bias, machine psychology

DATA AVAILABILITY STATEMENT

Datasets and analysis scripts are available in OSF (`https://osf.io/p8c39/files/2e58c42f-a7ba-45f2-aa60-265e107e36db`) and will be made public upon publication.

CONFLICT OF INTEREST

The authors declare no conflicts of interest.

ETHICS APPROVAL STATEMENT

This study did not involve human participants, animals, or identifiable personal data and therefore did not require ethics committee review.

PERMISSION TO REPRODUCE MATERIAL / CLINICAL TRIAL REGISTRATION

Not applicable.

AUTHOR CONTRIBUTIONS

KK contributed to Conceptualization, Methodology, Software, Formal Analysis, Data Curation, Writing – Original Draft, Visualization, Project Administration, and Funding Acquisition. YF, HG, TH, MN, YS, and IS contributed to Validation, Investigation, and Writing – Review & Editing. All authors after the first are listed alphabetically.

The field of artificial intelligence (AI) has undergone a profound transformation with the emergence of large language models (LLMs), marking a paradigm shift from rule-based systems to generative AI capable of producing human-like text (Ke, Tong, Cheng, & Peng, 2024). This evolution represents the culmination of decades of AI research, transitioning from early symbolic approaches to the current era of deep learning-based generative models with significant implications for psychological research and practice.

| From Symbolic AI to Generative AI

The history of AI formally began in 1956, with early developments featuring rule-based dialogue systems such as ELIZA (Rajaraman, 2023) and expert systems that encoded human expertise through explicit rules (Tan et

al., 2016). These symbolic approaches, grounded in semantic networks and ontologies (Shu-Hsien Liao, 2005; Sowa, n.d.), represented early attempts to manually engineer intelligence into computational systems.

The advent of machine learning and deep learning marked a fundamental shift from human-designed rules to data-driven optimization. This transformation reached a critical milestone with the Transformer architecture (Vaswani et al., 2023), which revolutionized natural language processing through self-attention mechanisms. Trained on massive text corpora (Gao et al., 2020), Transformer-based models evolved into large language models (LLMs) exemplified by OpenAI's GPT series, Anthropic's Claude (Enis & Hopkins, 2024), and Google's LaMDA (Thoppilan et al., 2022).

These LLMs constitute the foundation of modern generative AI, autonomously producing coherent and contextually appropriate natural language outputs. Their sophistication has sparked discussions about artificial general intelligence, positioning LLMs as central to contemporary AI discourse (Ferraris, Audrito, Caro, & Poncibò, 2025; Xi et al., 2023). This technological breakthrough presents unprecedented opportunities for psychological research and mental health applications.

LLMs in Psychological Research and Practice

The intersection of LLMs and psychology has generated multidisciplinary research streams. Researchers increasingly treat LLMs as subjects of psychological investigation (Abdurahman et al., 2024), examining their ethical reasoning patterns (Almeida, Nunes, Engelmann, Wiegmann, & Araújo, 2024) and their influence on human decision-making processes (Eigner & Händler, 2024). The development of psychology-specific LLMs (Hu et al., 2025) and the emergence of Machine Psychology—a field dedicated to understanding AI behavior through psychological frameworks (Hagendorff et al., 2023)—highlight the bidirectional relationship between psychology and artificial intelligence.

Mental healthcare represents a particularly promising application domain for LLM-based generative AI. Preliminary evidence suggests that AI counseling tools can enhance screening efficiency and alleviate professional workload (Stade et al., 2024). Nevertheless, significant concerns persist regarding inappropriate responses, algorithmic opacity, and potential biases (Demszky et al., 2023). Beyond serving as counselors, LLMs show potential as training clients (Maurya, 2024b) and performance evaluators (Qiu & Lan, 2024), offering comprehensive transformation possibilities for mental health education and practice.

Cultural and Linguistic Considerations

Despite the global reach of LLMs, their performance varies substantially across languages and cultural contexts (Zhong, Yun, & Sun, 2024). Current research remains disproportionately English-centric, creating critical gaps in our understanding of LLM capabilities in non-English contexts. This limitation proves especially problematic for mental health applications, where cultural sensitivity and linguistic nuance are fundamental to therapeutic efficacy. The evaluation of these systems in Japanese remains notably underexplored, despite Japan's distinct cultural frameworks for mental health and counseling practices that may influence AI performance and acceptability.

The Present Study

This research addresses these critical gaps by conducting the first comprehensive evaluation of LLM performance across three counseling roles in Japanese-language therapeutic contexts. Our investigation systematically examines the capabilities and limitations of state-of-the-art LLMs when deployed as counselors, clients, and evaluators in Japanese mental health scenarios.

We employed cutting-edge models available at the time of data collection. Counselor AIs included OpenAI's

GPT-4-turbo (implemented with either zero-shot prompting or Structured Multi-step Dialogue Prompts [SMDP]) and Anthropic's Claude-3-Opus (with SMDP). Claude-3-Opus served as the client AI, selected for its superior complex-dialogue capabilities despite computational costs (Berrezueta-Guzman, Kandil, Martín-Ruiz, {de la} Cruz, & Krusche, 2024), configured with demographically specified work-related concern prompts. Evaluation AIs comprised the latest generation models: OpenAI's o3, Anthropic's Sonnet, and Google's Gemini.

Our evaluation framework utilizes the Motivational Interviewing Treatment Integrity (MITI) Coding Manual 4.2.1 (Moyers, Manuel, & Ernst, 2014), consistent with established AI counseling research protocols (Cohen, Zisquit, Yosef, Friedman, & Bar, 2024; Steenstra, Nouraei, Arjmand, & Bickmore, 2024; Yang et al., 2025). This standardized approach enables rigorous comparison between AI and human evaluators while maintaining ecological validity for real-world therapeutic applications.

Research Questions and Hypotheses

Building on existing literature demonstrating the impact of prompting strategies on LLM counseling performance (Huang, Fu, Yang, Zhang, & Yang, 2024; Lee et al., 2024), the potential of LLMs as simulated clients (Gray et al., 2024; Potter & Jefferies, 2024), and the promise of automated evaluation systems (Rudolph, Seer, Mothes, & Albrecht, 2024; Sharma, Lin, Miner, Atkins, & Althoff, 2023), we formulated the following hypotheses:

H1: Counselor AIs utilizing SMDP will receive higher human-expert ratings than those using zero-shot prompting.
H2: Claude-3-Opus with SMDP will outperform GPT-4-turbo with SMDP on human-expert evaluations.
H3: Evaluations by o3, Sonnet, and Gemini-based systems will differ significantly from human-expert ratings.

This study advances Machine Psychology by examining practical implementation challenges of AI in mental health services. Through our focus on Japanese-language interactions, we provide essential evidence regarding cross-cultural validity and identify critical developmental priorities for non-English therapeutic contexts. Our findings have implications for the global deployment of AI-assisted mental health interventions and the development of culturally sensitive therapeutic AI systems.

METHOD

Counseling Script Development

We evaluated dialogue scripts generated through AI counselor–client interactions, subsequently assessed by human experts and AI evaluation systems. Three LLMs served as counselor AIs: (1) OpenAI's GPT–4–turbo with zero–shot prompting (GPT-zero); (2) GPT–4–turbo with Structured Multi-step Dialogue Prompts (GPT-SMDP); and (3) Anthropic's Claude–3–Opus–2000 with SMDP (Claude-SMDP). The SMDP framework facilitates secure communication environments through structured dialogue progression: current challenges → desired future state → attempted solutions → available resources → next steps, emphasizing reflective listening techniques (complete implementation details available in Appendix 1).

Client AIs used Claude–3–Opus–2000 with work–related concern prompts, specifying six profiles: three age groups (20s, 40s, 60s) × two sexes. A custom program maintained dialogue history via respective APIs. All scripts were generated on April 12, 2024 (complete prompts available in Appendix 2).

Human Expert Evaluation

Evaluators

Fifteen counseling experts evaluated the AI-generated scripts. Seventy percent had >10 years counseling experience; over half held Japanese clinical psychologist certification. The panel included career counselors, medical doctors, and other mental health professionals (Table 1).

Measures and Procedure

Evaluators assessed five indicators: four MITI coding system measuresCultivating Change Talk (CCT), Softening Sustain Talk (SST), Partnership (PAR), and Empathy (EMP)and one custom metric, Overall evaluation (OVR). Each evaluator received three blinded scripts (GPT-zero, GPT-SMDP, Opus-SMDP) from specific client AI profiles for simultaneous evaluation. Ratings employed the MITI 9-point scale (15 in 0.5 increments). Evaluators also rated client naturalness (010 scale) and provided qualitative feedback.

[Table 1 about here]

AI Evaluation Systems

Three AI systems evaluated counseling quality: OpenAI's o3, Anthropic's Claude–3.7–Sonnet, and Google's Gemini–2.5–pro. These systems followed the human evaluation protocol, rating the five indicators on the same 9-point scale (complete evaluation prompts and scoring rubrics available in Appendix 3).

Statistical Analysis

Counselor AI Performance

Linear mixed-effects models examined human ratings of counselor AI responses:

$$\begin{aligned}\mathsf{Score}_{ij} = \beta_0 &+ \beta_1\,\mathsf{Model}_{\text{one-shot},ij} + \beta_2\,\mathsf{Model}_{\text{gpt-smdp},ij} \\ &+ \beta_3\,\mathsf{AgeCat}_{40\text{s},ij} + \beta_4\,\mathsf{AgeCat}_{60\text{s},ij} + \beta_5\,\mathsf{Sex}_{\text{male},ij} \\ &+ u_{0j} + \varepsilon_{ij},\end{aligned} \tag{1}$$

where u_{0j} represents rater j's random intercept. Each indicator was modeled separately with Opus–SMDP/20s/female as reference. We conducted t-tests using Kenward–Roger/Satterthwaite degrees of freedom and three Tukey–Kramer adjusted pairwise comparisons.

AI Evaluation System Performance

Linear mixed-effects models assessed evaluation differences between humans and AI systems, including counselor model × evaluator group interactions:

$$\begin{aligned}\mathsf{Score}_{ijk} = \beta_0 &+ \beta_{1-2}\,\mathsf{Model}_{ij} + \beta_{3-5}\,\mathsf{Group}_{ij} + \beta_{6-11}\,(\mathsf{Model}\times\mathsf{Group})_{ij} \\ &+ \beta_{12-13}\,\mathsf{AgeCat}_{ij} + \beta_{14}\,\mathsf{Sex}_{ij} + u_{0j}^{(\text{human})} + v_{0k}^{(\text{AI})} + \varepsilon_{ijk},\end{aligned} \tag{2}$$

Random intercepts $u_{0j}^{(\text{human})}$ and $v_{0k}^{(\text{AI})}$ represent human and AI evaluator clusters. ML estimation with Satterthwaite approximation was employed. Six Tukey–Kramer adjusted comparisons examined main effects. Exploratory simple effects analyses used Cohen's d standardized by residual SD.

Inter-rater Reliability

ICC(2,1) and ICC(2,k) were calculated using R `psych` following Shrout and Fleiss's Model 2. Missing data were handled through pairwise deletion. Interpretation followed Koo & Li (2016): <0.50 poor, 0.50–0.75 moderate, 0.75–0.90 good, >0.90 excellent.

Sample Size and Pre-registration

Sample size calculations (G*Power 3.1) targeted 22 evaluations per comparison (medium effect = 0.5, power = 0.95, $\alpha = 0.025$ with Tukey adjustment), requiring 198 total evaluations. The final dataset contained 162 evaluations from 15 evaluators. Post-hoc simulations confirmed adequate power (Appendix 5).

Analyses used R v4.4.1 with packages lme4 v1.1.37, emmeans v1.11.0, and SIMR v1.0.7. The protocol was pre-registered at `https://doi.org/10.17605/OSF.IO/VU286`.

Ethical Considerations

This study involved no human participants, animals, or identifiable data; ethics review was not required.

RESULTS

Descriptive Statistics

Table 2 presents descriptive statistics by evaluator group. Human experts completed 54 evaluations, while evaluation AIs completed 66 evaluations per model. Each evaluation included assessments of three counselor AI models. Client AI characteristics were balanced across age and gender. Human ratings of client AI naturalness averaged 4.72 ś 2.02 (10-point scale), indicating below-midpoint ratings with substantial variability. Human expert ratings ranged from 2.0 to 3.5 points across counselor AI models, whereas evaluation AIs provided consistently higher ratings with minimum scores around 2.5 points. Gemini produced the highest ratings, reaching 5.0 points on some measures.

[Table 2 about here]

Human Evaluation of Counselor AIs

GPT-zero scored significantly lower than Opus-SMDP across all measures (Tables 35). Differences ranged from -0.90 to -1.04 points for CCT, PAR, EMP, and OVR, with a smaller difference for SST (-0.50 points). GPT-zero also scored significantly lower than GPT-SMDP on all indices, with differences of -0.84 to -1.23 points for CCT/PAR/EMP/OVR and -0.34 points for SST. Opus-SMDP and GPT-SMDP did not differ significantly. Client AI age and gender showed no significant effects.

[Table 3 about here]
[Table 4 about here]
[Table 5 about here]

Counseling Evaluation AIs

Linear mixed-effects models showed that all AI evaluators rated SST and OVR significantly higher than humans (Tables 68). For PAR, Sonnet and Geminibut not o3exceeded human ratings; for EMP, only Sonnet did. No differences emerged for CCT (Gemini: $p = .056$).

Client age negatively predicted PAR and EMP ratings, while male clients received higher scores on CCT, PAR, EMP, and OVR.

Pairwise comparisons revealed that Gemini produced the highest OVR ratings, followed by Sonnet and o3. For SST, PAR, and EMP, Sonnet and Gemini did not differ but both exceeded o3. For CCT, Gemini surpassed both Sonnet and o3, which did not differ from each other.

[Table 9 about here]

Inter-rater Reliability

Human evaluators showed good single-measure reliability for PAR and SST (ICCs > 0.79), but poor reliability for CCT and OVR (ICCs < 0.35). Average-measure reliability was excellent for most metrics (ICCs > 0.90) except CCT and OVR.

AI evaluator reliability varied markedly across modelmetric combinations. Some combinations yielded insufficient variance for ICC computation, while others achieved excellent average-measure reliability (0.90), particularly for o3 across multiple metrics and Sonnet for CCT, SST, and OVR. Single-measure reliability remained generally low across AI evaluators.

[Table 9 about here]

Qualitative Evaluation

Qualitative findings are detailed in Appendix 7.

Counselor AI Evaluation

Human experts identified four common limitations across AI counselors: excessive directiveness and advice-giving, insufficient emotional empathy and exploration of unexpressed content, inadequate deepening of change-oriented statements, and an evaluative stance hindering collaboration. Opus-SMDP effectively elicited client strengths but dominated sessions through excessive verbalization, resulting in ambiguous goal-setting and protracted closings. GPT-zero demonstrated rapid support but rushed toward problem identification and termination without adequate exploration, relying on generic encouragement. GPT-SMDP exhibited collaborative intent but employed monotonous, mechanical patterns that failed to explore client values or workplace challenges.

AI evaluators identified superficial change talk, surface-level empathy, premature advice-giving, and absence of complex reflections, with power-sharing remaining moderate. However, each AI evaluator emphasized different aspects: o3 focused on verbalization quantity and technical proficiency, Sonnet prioritized client affirmation and self-efficacy, while Gemini emphasized client autonomy and empowerment.

Client AI Evaluation

Human experts identified three primary limitations in client AI performance. First, dialogue patterns included unconditional acceptance of suggestions, mechanical repetition, flat interactions, translation artifacts, excessively

formal expressions, and awkward endings with formulaic phrases. Second, emotional expression was characterized by passive compliance, minimal resistance or ambivalence, and absent negative emotions. Third, character development lacked specificity, with abstract problem descriptions, missing age-appropriate emotions, and insufficient workplace or family context. Recommendations for improvement include incorporating natural dialogue elements (backchanneling, silences, varied responses), enhancing emotional range and bidirectionality, and adding specific contextual details to increase realism.

DISCUSSION

This study provides the first comprehensive evaluation of LLM performance across three counseling roles—counselor, client, and evaluator—in Japanese-language therapeutic contexts. Our findings reveal both the promise and limitations of current AI systems for mental health applications.

Hypothesis Testing Results

H1 (Structured prompting effects)

As predicted, counselor AIs using SMDP significantly outperformed zero-shot prompting across all human-rated metrics (increases of approximately 0.91.0 points). The structured dialogue cycle (Listen–Back 1 → Listen–Back 2 → Question) enhanced reflective responding, client-aligned questioning, and appropriate pacing before advice-giving. This demonstrates that explicit procedural guidance substantially improves AI counseling quality beyond simple role instructions.

H2 (Model differences)

GPT-4-turbo and Claude-3-Opus showed statistically equivalent performance when using SMDP (OVR difference = 0.06, $p = 0.624$), indicating comparable Japanese conversational capabilities. Power analysis suggests that a sample of at least 375 evaluators would be required to confirm true equivalence.

H3 (HumanAI evaluator differences)

Evaluation AIs systematically overrated counselor performance on Overall Quality (OVR). For Session Structure and Time Management (SST), leniency was evident for Sonnet and Gemini, whereas o3 did not differ from human ratings; for CCT no significant differences were observed. No significant differences were detected between AI and human ratings on PAR or EMP. Qualitative inspection revealed model-specific tendencies: Gemini emphasized power-sharing, o3 focused on technical proficiency, and Sonnet prioritized emotional expression. These findings highlight persisting challenges in automated counseling assessment.

Enhancing Counselor AI Performance

Despite the gains from SMDP, several limitations persisted: superficial reflections lacking emotional depth, rigid adherence to predetermined question flows, and premature or poorly timed advice-giving. Although SMDP respected client autonomy to a minimal degree, it still failed to facilitate genuine emotional engagement or deeper dialogue.

Strategic improvements should target reflective depth via Retrieval-Augmented Generation that captures implicit emotions and values (Arabi, Koyuturk, O'Mahony, Calati, & Ognibene, 2024). Dynamic flow management could be implemented through multi-agent architectures in which specialized LLMs handle strategic

control and utterance generation separately (Kampman et al., 2024; Xu et al., 2025). Advice timing may be improved by incorporating motivational-interviewing techniques such as AskOfferAsk, and pacing control through strategic pauses and response-length limits would allow adequate client reflection time.

Developing Realistic Client AIs

Human evaluators identified critical deficits in client-AI realism: flat affect, excessive compliance, and premature session termination. These shortcomings stem from limited persona depth, monotonous discourse patterns, and superficial scenario construction (Maurya, 2024a; Qiu & Lan, 2024).

Effective solutions include dynamic emotion modeling with resistance parameters (Bao, Liu, Huang, & Wei, 2025) and variable response patterns of one to three sentences with natural hesitations and back-channeling. Challenge-oriented scenarios that follow realistic counseling trajectories (Du et al., 2024), combined with parameterized goals and fears expressed throughout the conversation (Cook, Overgaard, Pankratz, Fiol, & Aakre, 2025), would substantially enhance realism.

Improving Evaluation AI Systems

Evaluation AIs exhibited high internal consistency yet systematic leniency relative to human experts, with each model emphasizing different counseling dimensions (power-sharing, technical skills, or emotional attunement).

Two factors appear to explain these discrepancies. First, surface-level bias led AI evaluators to reward politeness and positive language while missing nuanced deficits such as premature advice or weak collaboration, inflating OVR and, for some models, SST. Second, MITI interpretation challenges arose when global rating criteria (e.g., accepting/non-judgmental) were treated as achievable through courteous phrasing, while deeper requirements such as recognizing unspoken emotions were overlooked.

Improvement strategies should include providing full MITI manuals as context, developing LLM-optimized rubrics with explicit behavioral anchors, implementing chunk-based evaluation with subsequent integration, and fine-tuning on expert-rated exemplars using methods such as LoRA (Zhang et al., 2024).

Reliability Considerations

Inter-rater reliability analyses revealed important patterns. Human evaluators showed high agreement for SST and PAR (ICC > 0.79) but substantial variability for CCT and EMP, requiring multiple raters for reliable assessment (Koo & Li, 2016). Evaluation AIs demonstrated insufficient variance, suggesting leniency bias limits their discriminative capacity. Future systems should incorporate multiple foundation models to ensure adequate variance while maintaining consistency (Morgan, 2025).

Limitations

This study provides the first comprehensive evaluation of AI performance across counselor, client, and evaluation roles in Japanese counseling contexts. Several limitations warrant consideration.

First, our LLMs (GPT-4-turbo, Claude-3-Opus) became outdated between data collection and publicationa common challenge in AI research. While SMDPs demonstrated effectiveness likely generalizes to current models, verification is needed. Additionally, counselor AI excluded Googles Gemini owing to performance limitations at the time, although current versions may prove effective.

Second, we evaluated AI-to-AI dialogues rather than human-AI interactions. Whether counselor AI responds appropriately to human clients or client AI effectively trains human counselors remains unverified. Critically, AI performance with inputs deviating from prompt-specified flowscommon in real counselingwas

not examined. Ethical considerations, including transparency, data security, and AIs supplementary (not replacement) role, also require attention before implementation (Kiuchi et al., 2025).

Third, our MITI-based evaluation captures only Motivational Interviewing approaches, not all counseling modalities. MITIs design for audio recordings raises validity concerns for text evaluation. Furthermore, our findings are specific to Japanese-language contexts; documented language-dependent variations in LLMs limit cross-cultural generalizability.

CONCLUSION

This study presents the first comprehensive evaluation of counselor AI (GPT-4-turbo with zero-shot prompting, GPT-4-turbo with SMDP, Claude-3-Opus with SMDP), client AI, and counseling evaluation AI (o3, Claude-3.7-Sonnet, Gemini-2.5-pro) in Japanese counseling dialogues. SMDP-enhanced counselor AI significantly outperformed zero-shot conditions on MITI global ratings. GPT-4-turbo and Claude-3-Opus demonstrated comparable Japanese counseling quality. Evaluation AIs showed no significant differences from humans in CCT, PAR, or EMP ratings but exhibited marked leniency in OVR; for SST, significant leniency emerged only for Sonnet and Gemini, whereas o3 did not differ from human ratings. Model-specific biases were evident: Gemini emphasized power-sharing, o3 focused on technical skills, and Sonnet prioritized emotional expression. Client AI produced flat responses lacking resistance and emotional expression.

These findings suggest improvements through detailed persona settings with emotional parameters, model optimization via RAG and fine-tuning, and integrated advancement of prompt design, model selection, and evaluation frameworks. Our results inform research, development, and practical applications for Japanese counseling support and training. Future research should extend to human-AI interactions involving actual clients and counseling professionals.

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CONFLICT OF INTEREST

The authors declare no conflicts of interest.

SUPPORTING INFORMATION

The file `Supplemental_materials.pdf` contains the following items.

- Appendix 1. Counselor AI Prompt: the complete counsellor prompt and question sequence.
- Appendix 2. Client AI Prompt: a concise template used to generate the simulated client role.
- Appendix 3. LLM Evaluation Prompt: scoring rubrics and response format for the five metrics (CCT, SST, PAR, EMP, and OVR).
- Appendix 4. Counseling Script Examples: age-specific dialogue examples generated by GPT-4 turbo, Claude 3 Opus, and others.
- Appendix 5. Post-hoc Sample Size Design and Simulation: power analysis and simulation results, including Appendix Table 1.
- Appendix 6. Simple-Effect Contrasts for the Model Œ Evaluator-Group Interaction: tables of simple-effect comparisons for all metrics.

- Appendix 7. Qualitative Evaluation: qualitative comments provided by both human and LLM raters.

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TABLE 1 CHARACTERISTICS OF HUMAN RATERS

Characteristic	Frequency (%)
Counseling Experience	
$3 \leq$ yrs < 5	7 (13)
< 10 yrs	9 (17)
10+ yrs	38 (70)
Certificate	
Clinical Psychologist	30 (56)
Medical Doctor	7 (13)
Nurse	6 (11)
Occupational Therapist	7 (13)
Psychiatric Social Worker	8 (15)
Career Counselor	10 (19)

TABLE 2 COUNSELOR-AI SCORES BY EVALUATOR

Variable	Human (N=57)	Sonnet (N=66)	o3 (N=66)	Gemini (N=66)
Score for Opus-SMDP				
CCT	3.44 ± 0.87	3.57 ± 0.53	3.58 ± 0.44	4.43 ± 0.57
SST	3.38 ± 0.85	4.96 ± 0.13	4.42 ± 0.29	4.99 ± 0.06
PAR	2.99 ± 0.96	4.39 ± 0.21	3.72 ± 0.33	4.41 ± 0.55
EMP	3.22 ± 0.96	4.40 ± 0.29	3.94 ± 0.16	4.17 ± 0.27
OVR	3.28 ± 0.83	4.96 ± 0.13	4.83 ± 0.24	5.00 ± 0.00
Score for GPT-zero				
CCT	2.51 ± 0.95	2.55 ± 0.44	2.42 ± 0.33	3.21 ± 0.74
SST	2.88 ± 0.85	5.00 ± 0.00	4.22 ± 0.25	4.97 ± 0.12
PAR	2.08 ± 0.89	3.11 ± 0.64	2.71 ± 0.33	2.58 ± 0.63
EMP	2.18 ± 0.76	3.94 ± 0.44	3.44 ± 0.48	3.95 ± 0.25
OVR	2.38 ± 0.64	4.92 ± 0.26	4.64 ± 0.38	5.00 ± 0.00
Score for GPT-SMDP				
CCT	3.41 ± 0.74	3.46 ± 0.69	3.86 ± 0.26	3.92 ± 0.53
SST	3.22 ± 0.85	4.71 ± 0.25	4.08 ± 0.26	4.88 ± 0.28
PAR	3.20 ± 0.79	4.22 ± 0.31	4.08 ± 0.19	4.63 ± 0.42
EMP	3.12 ± 0.91	3.67 ± 0.30	3.80 ± 0.25	4.00 ± 0.00
OVR	3.22 ± 0.65	4.44 ± 0.16	3.98 ± 0.26	5.00 ± 0.00

Note: Client-AI characteristicsHuman evaluator group: age 20s 33%, 40s 32%, 60s 35%; female 51%; Naturalness 4.72 ± 2.02. For Sonnet, o3, and Gemini evaluator groups, the age distribution was 33% in each decade (20s/40s/60s) and the proportion of females was 50%.

TABLE 3. FIXED-EFFECTS ESTIMATES FOR THE FIVE MITI-BASED SCORES

Score	Term	Estimate	Std. Error	df	t value	p value
CCT						
	(Intercept)	3.39	0.17	44.78	19.38	<0.001
	model: GPT-zero	-0.93	0.12	158.18	-7.62	<0.001
	model: GPT-SMDP	-0.03	0.12	158.18	-0.22	0.829
	client age: 40s	-0.13	0.13	164.12	-1.02	0.311
	client age: 60s	0.12	0.12	161.03	0.95	0.341
	client sex: male	0.08	0.10	163.57	0.73	0.465
SST						
	(Intercept)	3.34	0.17	64.81	19.65	<0.001
	model: GPT-zero	-0.50	0.14	158.73	-3.68	<0.001
	model: GPT-SMDP	-0.16	0.14	158.73	-1.16	0.247
	client age: 40s	-0.10	0.14	165.70	-0.70	0.486
	client age: 60s	0.04	0.14	162.26	0.27	0.787
	client sex: male	0.12	0.11	165.08	1.07	0.287
PAR						
	(Intercept)	2.98	0.18	57.95	16.61	<0.001
	model: GPT-zero	-0.91	0.14	158.38	-6.57	<0.001
	model: GPT-SMDP	0.21	0.14	158.38	1.52	0.131
	client age: 40s	-0.08	0.14	165.23	-0.58	0.562
	client age: 60s	-0.11	0.14	161.79	-0.79	0.433
	client sex: male	0.22	0.12	164.61	1.88	0.062
EMP						
	(Intercept)	3.21	0.18	63.57	17.87	<0.001
	model: GPT-zero	-1.04	0.15	157.53	-7.15	<0.001
	model: GPT-SMDP	-0.10	0.15	157.53	-0.66	0.510
	client age: 40s	-0.02	0.15	165.30	-0.12	0.905
	client age: 60s	0.01	0.15	161.48	0.07	0.943
	client sex: male	0.16	0.12	164.61	1.34	0.182
OVR						
	(Intercept)	3.30	0.14	95.81	23.80	<0.001
	model: GPT-zero	-0.90	0.13	159.13	-7.22	<0.001
	model: GPT-SMDP	-0.06	0.13	159.13	-0.49	0.624
	client age: 40s	-0.08	0.13	166.66	-0.62	0.535
	client age: 60s	-0.03	0.12	163.27	-0.25	0.805
	client sex: male	0.04	0.10	166.09	0.37	0.712

Note: The intercept corresponds to the baseline condition (model = Opus-SMDP, client age = 20 s, client sex = female). GPT-zero and GPT-SMDP are the two comparison models.

TABLE 4. TUKEY-ADJUSTED PAIRWISE CONTRASTS AMONG COUNSELOR MODELS

Contrast	Score	Estimate	Std. Error	df	t value	p value
Opus-SMDP − GPT-zero						
	CCT	0.93	0.12	161.44	7.50	<0.001
	SST	0.50	0.14	161.75	3.62	0.001
	PAR	0.91	0.14	161.63	6.47	<0.001
	EMP	1.04	0.15	161.68	7.04	<0.001
	OVR	0.90	0.13	162.35	7.11	<0.001
Opus-SMDP − GPT-SMDP						
	CCT	0.03	0.12	161.44	0.21	0.975
	SST	0.16	0.14	161.75	1.14	0.490
	PAR	-0.21	0.14	161.63	-1.49	0.297
	EMP	0.10	0.15	161.68	0.65	0.792
	OVR	0.06	0.13	162.35	0.48	0.879
GPT-zero − GPT-SMDP						
	CCT	-0.90	0.12	161.44	-7.29	<0.001
	SST	-0.34	0.14	161.75	-2.47	0.038
	PAR	-1.12	0.14	161.63	-7.96	<0.001
	EMP	-0.95	0.15	161.68	-6.39	<0.001
	OVR	-0.84	0.13	162.35	-6.62	<0.001

Note: Estimates are Tukeyadjusted pairwise contrasts; positive values favor the first model in each contrast.

TABLE 5. RANDOM-EFFECTS VARIANCE COMPONENTS (RATER INTERCEPTS & RESIDUALS)

Score	Group	Term	Variance	SD
CCT	Evaluator	Intercept	0.47	0.68
	Residual	Observation	0.65	0.81
SST	Evaluator	Intercept	0.37	0.61
	Residual	Observation	0.73	0.85
PAR	Evaluator	Intercept	0.42	0.65
	Residual	Observation	0.74	0.86
EMP	Evaluator	Intercept	0.38	0.62
	Residual	Observation	0.78	0.88
OVR	Evaluator	Intercept	0.22	0.47
	Residual	Observation	0.67	0.82

Note: Variance is the randomeffects variance, and SD is its square root. Rows labeled Residual represent withinrater observationlevel error.

TABLE 6. FIXED-EFFECT ESTIMATES FROM THE LINEAR MIXED-EFFECTS MODELS

Score	Term	Estimate	Std. Error	df	t value	p value
CCT						
	(Intercept)	3.28	0.15	28.58	22.54	<0.001
	model: GPT-zero	-0.93	0.10	750.36	-9.11	<0.001
	model: GPT-SMDP	-0.03	0.10	750.36	-0.26	0.797
	evaluator group: Sonnet	0.15	0.48	14.16	0.30	0.766
	evaluator group: o3	0.16	0.48	14.16	0.33	0.743
	evaluator group: Gemini	1.01	0.48	14.16	2.08	0.056
	client age: 40s	0.06	0.05	755.44	1.17	0.241
	client age: 60s	0.12	0.05	752.84	2.46	0.014
	client sex: male	0.17	0.04	755.06	4.21	<0.001
	model(GPT-zero) × evaluator group(Sonnet)	-0.09	0.14	750.36	-0.61	0.540
	model(GPT-SMDP) × evaluator group(Sonnet)	-0.08	0.14	750.36	-0.57	0.567
	model(GPT-zero) × evaluator group(o3)	-0.23	0.14	750.36	-1.65	0.100
	model(GPT-SMDP) × evaluator group(o3)	0.31	0.14	750.36	2.20	0.028
	model(GPT-zero) × evaluator group(Gemini)	-0.29	0.14	750.36	-2.08	0.038
	model(GPT-SMDP) × evaluator group(Gemini)	-0.49	0.14	750.36	-3.51	<0.001
SST						
	(Intercept)	3.38	0.12	23.73	28.92	<0.001
	model: GPT-zero	-0.50	0.07	749.44	-7.01	<0.001
	model: GPT-SMDP	-0.16	0.07	749.44	-2.21	0.027
	evaluator group: Sonnet	1.59	0.41	14.12	3.90	0.002
	evaluator group: o3	1.06	0.41	14.12	2.58	0.022
	evaluator group: Gemini	1.62	0.41	14.12	3.97	0.001
	client age: 40s	-0.05	0.03	753.76	-1.60	0.110
	client age: 60s	0.03	0.03	751.50	0.76	0.448
	client sex: male	0.00	0.03	753.45	0.16	0.874
	model(GPT-zero) × evaluator group(Sonnet)	0.54	0.10	749.44	5.53	<0.001
	model(GPT-SMDP) × evaluator group(Sonnet)	-0.09	0.10	749.44	-0.95	0.344
	model(GPT-zero) × evaluator group(o3)	0.30	0.10	749.44	3.04	0.002
	model(GPT-SMDP) × evaluator group(o3)	-0.18	0.10	749.44	-1.88	0.060
	model(GPT-zero) × evaluator group(Gemini)	0.48	0.10	749.44	4.90	<0.001
	model(GPT-SMDP) × evaluator group(Gemini)	0.04	0.10	749.44	0.45	0.649
PAR						
	(Intercept)	3.09	0.13	28.21	23.23	<0.001
	model: GPT-zero	-0.91	0.09	750.17	-9.79	<0.001
	model: GPT-SMDP	0.21	0.09	750.17	2.26	0.024
	evaluator group: Sonnet	1.36	0.44	13.98	3.07	0.008
	evaluator group: o3	0.69	0.44	13.98	1.56	0.141
	evaluator group: Gemini	1.38	0.44	13.98	3.12	0.008
	client age: 40s	-0.26	0.04	755.31	-5.74	<0.001
	client age: 60s	-0.15	0.04	752.68	-3.41	0.001
	client sex: male	0.15	0.04	754.93	4.19	<0.001
	model(GPT-zero) × evaluator group(Sonnet)	-0.37	0.13	750.17	-2.89	0.004
	model(GPT-SMDP) × evaluator group(Sonnet)	-0.38	0.13	750.17	-2.97	0.003
	model(GPT-zero) × evaluator group(o3)	-0.10	0.13	750.17	-0.75	0.454
	model(GPT-SMDP) × evaluator group(o3)	0.15	0.13	750.17	1.20	0.229
	model(GPT-zero) × evaluator group(Gemini)	-0.92	0.13	750.17	-7.24	<0.001
	model(GPT-SMDP) × evaluator group(Gemini)	0.01	0.13	750.17	0.07	0.943
EMP						
	(Intercept)	3.30	0.13	24.68	26.25	<0.001
	model: GPT-zero	-1.04	0.08	749.16	-12.70	<0.001
	model: GPT-SMDP	-0.10	0.08	749.16	-1.17	0.241
	evaluator group: Sonnet	1.09	0.43	13.51	2.53	0.025
	evaluator group: o3	0.63	0.43	13.51	1.46	0.168
	evaluator group: Gemini	0.85	0.43	13.51	1.98	0.068
	client age: 40s	-0.09	0.04	754.08	-2.21	0.028
	client age: 60s	-0.11	0.04	751.53	-2.72	0.007
	client sex: male	0.15	0.03	753.72	4.80	<0.001
	model(GPT-zero) × evaluator group(Sonnet)	0.58	0.11	749.16	5.18	<0.001

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Score	Term	Estimate	Std. Error	df	t value	p value
	model(GPT-SMDP) × evaluator group(Sonnet)	-0.64	0.11	749.16	-5.69	<0.001
	model(GPT-zero) × evaluator group(o3)	0.54	0.11	749.16	4.85	<0.001
	model(GPT-SMDP) × evaluator group(o3)	-0.04	0.11	749.16	-0.36	0.722
	model(GPT-zero) × evaluator group(Gemini)	0.82	0.11	749.16	7.34	<0.001
	model(GPT-SMDP) × evaluator group(Gemini)	-0.07	0.11	749.16	-0.63	0.532
OVR						
	(Intercept)	3.27	0.08	32.45	39.05	<0.001
	model: GPT-zero	-0.90	0.07	750.14	-13.70	<0.001
	model: GPT-SMDP	-0.06	0.07	750.14	-0.93	0.352
	evaluator group: Sonnet	1.67	0.26	12.98	6.35	<0.001
	evaluator group: o3	1.54	0.26	12.98	5.86	<0.001
	evaluator group: Gemini	1.71	0.26	12.98	6.50	<0.001
	client age: 40s	-0.05	0.03	756.24	-1.48	0.138
	client age: 60s	-0.05	0.03	753.17	-1.62	0.105
	client sex: male	0.11	0.03	755.78	4.11	<0.001
	model(GPT-zero) × evaluator group(Sonnet)	0.86	0.09	750.14	9.53	<0.001
	model(GPT-SMDP) × evaluator group(Sonnet)	-0.46	0.09	750.14	-5.12	<0.001
	model(GPT-zero) × evaluator group(o3)	0.71	0.09	750.14	7.85	<0.001
	model(GPT-SMDP) × evaluator group(o3)	-0.79	0.09	750.14	-8.74	<0.001
	model(GPT-zero) × evaluator group(Gemini)	0.90	0.09	750.14	10.04	<0.001
	model(GPT-SMDP) × evaluator group(Gemini)	0.06	0.09	750.14	0.68	0.495

TABLE 7. VARIANCE COMPONENTS (RANDOM EFFECTS) FROM THE LINEAR MIXED-EFFECTS MODELS

Score	Group	Term	Variance	SD
CCT	evaluator: LLM	(Intercept)	0.00	0.00
	evaluator: human	(Intercept)	0.46	0.68
	Residual	Observation	0.54	0.74
SST	evaluator: LLM	(Intercept)	0.00	0.00
	evaluator: human	(Intercept)	0.39	0.62
	Residual	Observation	0.38	0.62
PAR	evaluator: LLM	(Intercept)	0.00	0.00
	evaluator: human	(Intercept)	0.42	0.65
	Residual	Observation	0.50	0.71
EMP	evaluator: LLM	(Intercept)	0.00	0.00
	evaluator: human	(Intercept)	0.41	0.64
	Residual	Observation	0.44	0.66
OVR	evaluator: LLM	(Intercept)	0.00	0.00
	evaluator: human	(Intercept)	0.25	0.50
	Residual	Observation	0.35	0.59

Note: Variance is the estimated random-effects variance; SD is its square root. evaluator: LLM denotes ratings produced by large-language-model evaluators, whereas evaluator: human denotes ratings produced by human experts. Rows labelled Residual represent observation-level (within-rater) error.

TABLE 8. TUKEY-ADJUSTED PAIRWISE COMPARISONS AMONG EVALUATOR GROUPS (MAIN EFFECT)

Score	Comparison	Estimate	Std. Error	df	t ratio	p value
CCT						
	human Sonnet	−0.09	0.52	14.71	−0.18	0.998
	human o3	−0.19	0.52	14.71	−0.36	0.983
	human Gemini	−0.75	0.52	14.71	−1.46	0.486
	Sonnet o3	−0.10	0.06	31.07	−1.74	0.322
	Sonnet Gemini	−0.66	0.06	31.07	−11.94	<0.001
	o3 Gemini	−0.56	0.06	31.07	−10.20	<0.001
SST						
	human Sonnet	−1.74	0.44	15.78	−3.99	0.005
	human o3	−1.09	0.44	15.78	−2.51	0.098
	human Gemini	−1.80	0.44	15.78	−4.12	0.004
	Sonnet o3	0.65	0.04	31.06	16.82	<0.001
	Sonnet Gemini	−0.06	0.04	31.06	−1.44	0.485
	o3 Gemini	−0.70	0.04	31.06	−18.26	<0.001
PAR						
	human Sonnet	−1.11	0.47	14.76	−2.35	0.130
	human o3	−0.71	0.47	14.76	−1.51	0.458
	human Gemini	−1.08	0.47	14.76	−2.29	0.147
	Sonnet o3	0.40	0.05	31.07	7.91	<0.001
	Sonnet Gemini	0.03	0.05	31.07	0.65	0.914
	o3 Gemini	−0.37	0.05	31.07	−7.26	<0.001
EMP						
	human Sonnet	−1.07	0.46	15.49	−2.33	0.135
	human o3	−0.79	0.46	15.49	−1.73	0.344
	human Gemini	−1.11	0.46	15.49	−2.40	0.118
	Sonnet o3	0.28	0.04	31.06	6.19	<0.001
	Sonnet Gemini	−0.04	0.04	31.06	−0.79	0.856
	o3 Gemini	−0.31	0.04	31.06	−6.98	<0.001
OVR						
	human Sonnet	−1.81	0.28	13.85	−6.45	<0.001
	human o3	−1.52	0.28	13.85	−5.42	<0.001
	human Gemini	−2.03	0.28	13.85	−7.26	<0.001
	Sonnet o3	0.29	0.04	31.06	8.07	<0.001
	Sonnet Gemini	−0.23	0.04	31.06	−6.37	<0.001
	o3 Gemini	−0.52	0.04	31.06	−14.43	<0.001

Note: Estimates are Tukeyadjusted mean differences (first group second group). A positive estimate indicates the first evaluator group gave higher scores. p values correspond to Tukey multiplicityadjusted tests.

TABLE 9. INTER-RATER RELIABILITY (INTRACLASS CORRELATION COEFFICIENTS)

	ICC (2,1)[a]				ICC (2,k)[b]			
Score	Human	Sonnet	o3	Gemini	Human	Sonnet	o3	Gemini
CCT	0.33	0.57	0.89	0.38	0.75	0.94	0.99	0.87
SST	0.79	0.51	0.86	0.28	0.98	0.92	0.99	0.70
PAR	0.90	0.25	0.25	0.80	0.99	0.79	0.67	0.98
EMP	0.65	0.31	0.85	0.80	0.95	0.73	0.98	0.98
OVR	0.11	0.87	0.59	–	0.42	0.99	0.94	–

[a] Twoway random, singlemeasure consistency ICC.
[b] Twoway random, averagemeasure consistency ICC.
– indicates that the statistic could not be estimated (zero numerator variance).